

Name:Naveena C J

USN: 1AM22CD061

e-mail:Naveenacj99@gmail.com

lab 5:

```
data class Contact(
```

```
    val name: String,
```

```
    val phone: String?,
```

```
    val email: String?,
```

```
    val favorite: Boolean
```

```
)
```

```
fun formatContact(c: Contact, index: Int): String {
```

```
    val phoneDisplay = c.phone ?: "N/A"
```

```
    val emailDisplay = c.email ?: "N/A"
```

```
    val star = if (c.favorite) " *" else ""
```

```
    return "$index. ${c.name} | Phone: $phoneDisplay | Email: $emailDisplay} |$star"
```

```
}
```

```
fun filterContacts(
```

```
    contacts: List<Contact>,
```

```
    predicate: (Contact) -> Boolean
```

```
): List<Contact> {
```

```
    return contacts.filter(predicate)
```

```
}
```

```
fun printContacts(title: String, contacts: List<Contact>) {
```

```
    println(title)
```

```

contacts.forEachIndexed { index, contact ->
    println(formatContact(contact, index + 1))
}
println()
}

fun main() {
    val allContacts = listOf(
        Contact("Maya Patel", "+91-90000-11111", "maya@example.com", true),
        Contact("Raj Sharma", null, "raj@sample.com", false),
        Contact("Li Wei", "+86-13800000000", null, false),
        Contact("Sara Ali", null, null, false),
        Contact("Tom O'Brien", "+1-555-1234", "tom@example.org", true)
    )

    printContacts("--- All Contacts ---", allContacts)

    printContacts(
        "--- Favorite Contacts ---",
        filterContacts(allContacts) { it.favorite }
    )

    printContacts(
        "--- Contacts with Email ---",
        filterContacts(allContacts) { it.email != null }
    )
}

```